

CHAPTER 3

MATERIAL BALANCE APPLIED TO OIL RESERVOIRS

3.1 INTRODUCTION

The Schilthuis material balance equation has long been regarded as one of the basic tools of reservoir engineers for interpreting and predicting reservoir performance.

In this chapter, the zero dimensional material balance is derived and subsequently applied, using mainly the interpretative technique of Havlena and Odeh, to gain an understanding of reservoir drive mechanisms under primary recovery conditions. Finally, some of the uncertainties attached to estimation of in-situ pore compressibility, a basic component in the material balance equation, are qualitatively discussed.

Although the classical material balance techniques, once applied, have now largely been superseded by numerical simulators, which are essentially multi-dimensional, multi-phase, dynamic material balance programs, the classical approach is well worth studying since it provides a valuable insight into the behaviour of hydrocarbon reservoirs.

3.2 GENERAL FORM OF THE MATERIAL BALANCE EQUATION FOR A HYDROCARBON RESERVOIR

The general form of the material balance equation was first presented by Schilthuis¹ in 1941. The equation is derived as a volume balance which equates the cumulative observed production, expressed as an underground withdrawal, to the expansion of the fluids in the reservoir resulting from a finite pressure drop. The situation is depicted in fig. 3.1 in which (a) represents the fluid volume at the initial pressure p_i in a reservoir which has a finite gascap. The total fluid volume in this diagram is the hydrocarbon pore volume of the reservoir (HCPV). Fig. 3.1 (b) illustrates the effect of reducing the pressure by an amount Δp and allowing the fluid volumes to expand, in an artificial sense, in the reservoir. The original HCPV is still drawn in this diagram as the solid line. Volume A is the increase due to the expansion of the oil plus originally dissolved gas, while volume increase B is due to the expansion of the initial gascap gas. The third volume increment C is the decrease in HCPV due to the combined effects of the expansion of the connate water and reduction in reservoir pore volume as already discussed in Chapter 1, sec. 7.

If the total observed surface production of oil and gas is expressed in terms of an underground withdrawal, evaluated at the lower pressure p , (which means effectively,



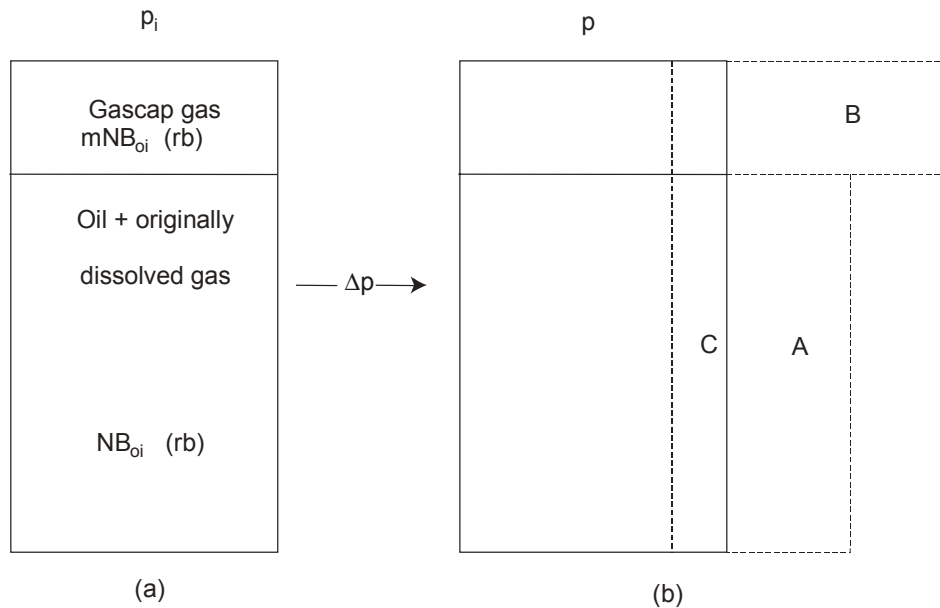


Fig. 3.1 Volume changes in the reservoir associated with a finite pressure drop Δp ; (a) volumes at initial pressure, (b) at the reduced pressure

taking all the surface production back down to the reservoir at this lower pressure) then it should fit into the volume $A + B + C$ which is the total volume change of the original HCPV. Conversely, volume $A + B + C$ results from expansions which are allowed to artificially occur in the reservoir. In reality, of course, these volume changes correspond to reservoir fluid which would be expelled from the reservoir as production. Thus the volume balance can be evaluated in reservoir barrels as

$$\begin{aligned}
 \text{Underground withdrawal (rb)} &= \text{Expansion of oil + originally dissolved gas (rb)} \\
 &+ \text{Expansion of gascap gas (rb)} \\
 &+ \text{Reduction in HCPV due to connate water expansion and decrease in the pore volume (rb)}
 \end{aligned}$$

Before evaluating the various components in the above equation it is first necessary to define the following parameters.

N is the initial oil in place in stock tank barrels
 $= V \phi (1 - S_{wc}) / B_{oi}$ stb

m is the ratio

$$\frac{\text{initial hydrocarbon volume of the gascap}}{\text{initial hydrocarbon volume of the oil}}$$

(and, being defined under initial conditions, is a constant)

N_p is the cumulative oil production in stock tank barrels, and

R_p is the cumulative gas oil ratio



$$= \frac{\text{Cumulative gas production (scf)}}{\text{Cumulative oil production (stb)}}$$

Then the expansion terms in the material balance equation can be evaluated as follows.

a) Expansion of oil plus originally dissolved gas

There are two components in this term:

- Liquid expansion

The N stb will occupy a reservoir volume of NB_{oi} rb, at the initial pressure, while at the lower pressure p , the reservoir volume occupied by the N stb will be NB_o , where B_o is the oil formation volume factor at the lower pressure. The difference gives the liquid expansion as

$$N(B_o - B_{oi}) \quad (\text{rb}) \quad (3.1)$$

- Liberated gas expansion

Since the initial oil is in equilibrium with a gascap, the oil must be at saturation or bubble point pressure. Reducing the pressure below p_i will result in the liberation of solution gas. The total amount of solution gas in the oil is NR_{si} scf. The amount still dissolved in the N stb of oil at the reduced pressure is NR_s scf. Therefore, the gas volume liberated during the pressure drop Δp , expressed in reservoir barrels at the lower pressure, is

$$N(R_{si} - R_s) B_g \quad (\text{rb}) \quad (3.2)$$

b) Expansion of the gascap gas

The total volume of gascap gas is mNB_{oi} rb, which in scf may be expressed as

$$G = \frac{mNB_{oi}}{B_{gi}} \quad (\text{scf})$$

This amount of gas, at the reduced pressure p , will occupy a reservoir volume

$$mNB_{oi} \frac{B_g}{B_{gi}} \quad (\text{rb})$$

Therefore, the expansion of the gascap is

$$mNB_{oi} \left(\frac{B_g}{B_{gi}} - 1 \right) \quad (\text{rb}) \quad (3.3)$$

c) Change in the HCPV due to the connate water expansion and pore volume reduction



The total volume change due to these combined effects can be mathematically expressed as

$$d(\text{HCPV}) = -dV_w + dV_f \quad (1.36)$$

or, as a reduction in the hydrocarbon pore volume, as

$$d(\text{HCPV}) = -(c_w V_w + c_f V_f) \Delta p \quad (1.38)$$

where V_f is the total pore volume = $\text{HCPV}/(1 - S_{wc})$

and V_w is the connate water volume = $V_f \times S_{wc} = (\text{HCPV})S_{wc}/(1 - S_{wc})$.

Since the total HCPV, including the gascap, is

$$(1+m)NB_{oi} \quad (\text{rb}) \quad (3.4)$$

then the HCPV reduction can be expressed as

$$-d(\text{HCPV}) = (1+m)NB_{oi} \left(\frac{c_w S_{wc} + c_f}{1 - S_{wc}} \right) \Delta p \quad (3.5)$$

This reduction in the volume which can be occupied by the hydrocarbons at the lower pressure, p , must correspond to an equivalent amount of fluid production expelled from the reservoir, and hence should be added to the fluid expansion terms.

d) Underground withdrawal

The observed surface production during the pressure drop Δp is N_p stb of oil and $N_p R_p$ scf of gas. When these volumes are taken down to the reservoir at the reduced pressure p , the volume of oil plus dissolved gas will be $N_p B_o$ rb. All that is known about the total gas production is that, at the lower pressure, $N_p R_s$ scf will be dissolved in the N_p stb of oil. The remaining produced gas, $N_p (R_p - R_s)$ scf is therefore, the total amount of liberated and gascap gas produced during the pressure drop Δp and will occupy a volume $N(R_p - R_s)B_g$ rb at the lower pressure. The total underground withdrawal term is therefore

$$N_p (B_o + (R_p - R_s)B_g) \quad (\text{rb}) \quad (3.6)$$

Therefore, equating this withdrawal to the sum of the volume changes in the reservoir, eqs. (3.1), (3.2), (3.3) and (3.5), gives the general expression for the material balance as

$$N_p (B_o + (R_p - R_s)B_g) = NB_{oi} \left[\frac{(B_o - B_{oi}) + (R_{si} - R_s)B_g}{B_{oi}} + m \left(\frac{B_g}{B_{gi}} - 1 \right) + (1+m) \left(\frac{c_w S_{wc} + c_f}{1 - S_{wc}} \right) \Delta p \right] + (W_e - W_p)B_w \quad (3.7)$$



in which the final term $(W_e - W_p)B_w$ is the net water influx into the reservoir. This has been intuitively added to the right hand side of the balance since any such influx must expel an equivalent amount of production from the reservoir thus increasing the left hand side of the equation by the same amount. In this influx term

W_e = Cumulative water influx from the aquifer into the reservoir, stb.

W_p = Cumulative amount of aquifer water produced, stb.

and B_w = Water formation volume factor rb/stb.

B_w is generally close to unity since the solubility of gas in water is rather small and this condition will be assumed throughout this text. For more detailed calculations, correlation charts for B_w are presented in references 2 and 3.

The following features should be noted in connection with the expanded material balance equation

- it is zero dimensional, meaning that it is evaluated at a point in the reservoir
- it generally exhibits a lack of time dependence although, as will be discussed in sec. 3.7 and also in Chapter 9, the water influx has a time dependence
- although the pressure only appears explicitly in the water and pore compressibility term as, $\Delta p = p_i - p$, it is implicit in all the other terms since the PVT parameters B_o , R_s and B_g are themselves functions of pressure. The water influx is also pressure dependent.
- the equation is always evaluated, in the way it was derived, by comparing the current volumes at pressure p to the original volumes at p_i . It is not evaluated in a step-wise or differential fashion.

Although the equation appears a little intimidating, at first sight, it should be thought of as nothing more than a sophisticated version of the compressibility definition

$$dV = c \times V \times \Delta p$$

Production = Expansion of reservoir fluids.

and, under certain circumstances, can in fact be reduced to this simple form.

In using the material balance equation, one of the main difficulties lies in the determination of the representative average reservoir pressure at which the pressure dependent parameters in the equation should be evaluated. This follows from the zero dimensional nature of the equation which implies that there should be some point in the reservoir at which a volume averaged pressure can be uniquely determined. In applying the more simple gas material balance, equ. (1.35), such a point could be defined with reasonable accuracy as the centroid point, at which pressures could be evaluated throughout the producing life of the



reservoir. In the case of an oil reservoir, however, the situation is generally more complex since below the bubble point two phases, oil and gas, will co-exist and, due to the gravity difference between the phases, will tend to segregate. As a result, the point at which the average pressure should be determined will vary with time. Precisely how the volume averaged reservoir pressure can be determined from the analysis of pressure tests in wells will be detailed in Chapter 7.

3.3 THE MATERIAL BALANCE EXPRESSED AS A LINEAR EQUATION

Since the advent of sophisticated numerical reservoir simulation techniques, the Schilthuis material balance equation has been regarded by many engineers as being of historical interest only; a technique used back in the nineteen forties and fifties when people still used slide-rules. It is therefore interesting to note that as late as 1963-4, Havlena and Odeh presented two of the most interesting papers ever published on the subject of applying the material balance equation and interpreting the results. Their papers,^{4,5} described the technique of interpreting the material balance as the equation of a straight line, the first paper describing the technique and the second illustrating the application to reservoir case histories.

To express equ. (3.7) in the way presented by Havlena and Odeh requires the definition of the following terms

$$F = N_p (B_o + (R_p - R_s) B_g) + W_p B_w \quad (\text{rb}) \quad (3.8)$$

which is the underground withdrawal;

$$E_o = (B_o - B_{oi}) + (R_{si} - R_s) B_g \quad (\text{rb/stb}) \quad (3.9)$$

which is the term describing the expansion of the oil and its originally dissolved gas;

$$E_g = B_{oi} \left(\frac{B_g}{B_{gi}} - 1 \right) \quad (\text{rb/stb}) \quad (3.10)$$

describing the expansion of the gascap gas, and

$$E_{f,w} = (1+m) B_{oi} \left(\frac{c_w S_{wc} + c_f}{1 - S_{wc}} \right) \Delta p \quad (\text{rb/stb}) \quad (3.11)$$

for the expansion of the connate water and reduction in the pore volume. Using these terms the material balance equation can be written as

$$F = N(E_o + mE_g + E_{f,w}) + W_e B_w \quad (3.12)$$

Havlena and Odeh have shown that in many cases equ. (3.12) can be interpreted as a linear function. For instance, in the case of a reservoir which has no initial gascap, negligible water influx and for which the connate water and rock compressibility term may be neglected; the equation can be reduced to



$$F = NE_o \quad (3.13)$$

in which the observed production, evaluated as an underground withdrawal, should plot as a linear function of the expansion of the oil plus its originally dissolved gas, the latter being calculated from a knowledge of the PVT parameters at the current reservoir pressure. This interpretation technique is useful, in that, if a simple linear relationship such as equ. (3.13) is expected for a reservoir and yet the actual plot turns out to be non-linear, then this deviation can itself be diagnostic in determining the actual drive mechanisms in the reservoir. For instance, equ. (3.13) may turn out to be non-linear because there is an unsuspected water influx into the reservoir helping to maintain the pressure. In this case equ. (3.12) can still be expressed in a linear form as

$$\frac{F}{E_o} = n + \frac{W_e}{E_o} \quad (3.14)$$

in which F/E_o should now plot as a linear function of W_e/E_o .

Once a straight line has been achieved, based on matching observed production and pressure data, then the engineer has, in effect, built a suitable mathematical model to describe the performance of the reservoir. As previously described, in Chapter 1, sec. 7, this phase is commonly referred to as a history match. Once this has been satisfactorily achieved, the next step is to use the same mathematical model to predict how the reservoir will perform in the future, possibly for a variety of production schemes. This prediction phase is facilitated by the mathematical ease in using the simple linear expressions for the material balance equation, as presented by Havlena and Odeh. The technique will be illustrated in greater detail in the following sections.

3.4 RESERVOIR DRIVE MECHANISMS

If none of the terms in the material balance equation can be neglected, then the reservoir can be described as having a combination drive in which all possible sources of energy contribute a significant part in producing the reservoir fluids and determining the primary recovery factor. In many cases, however, reservoirs can be singled out as having predominantly one main type of drive mechanism in comparison to which all other mechanisms have a negligible effect. In the following sections, such reservoirs will be described in order to isolate and study the contribution of the individual components in the material balance in influencing the recovery factor and determining the production policy of the field. The mechanisms which will be studied are:

- solution gas drive
- gascap drive
- natural water drive
- compaction drive

And these individual reservoir drive mechanisms will be investigated in terms of:

- reducing the material balance to a compact form, in many cases using the technique of Havlena and Odeh, in order to quantify reservoir performance



- determining the main producing characteristics, the producing gas oil ratio and watercut
- determining the pressure decline in the reservoir
- estimating the primary recovery factor
- investigating the possibilities of increasing the primary recovery.

3.5 SOLUTION GAS DRIVE

A solution gas drive reservoir is one in which the principal drive mechanism is the expansion of the oil and its originally dissolved gas. The increase in fluid volumes during the process is equivalent to the production.

Two phases can be distinguished, as shown in fig. 3.2 (a) when the reservoir oil is undersaturated and (b) when the pressure has fallen below the bubble point and a free gas phase exists in the reservoir.

a) Above bubble point pressure (undersaturated oil)

For a solution gas drive reservoir it is assumed that there is no initial gascap, thus $m = 0$, and that the aquifer is relatively small in volume and the water influx is negligible. Furthermore, above the bubble point, $R_s = R_{si} = R_p$, since all the gas produced at the surface must have been dissolved in the oil in the reservoir.

Under these assumptions, the material balance equation, (3.7), can be reduced to

$$N_p B_o = N B_{oi} \left(\frac{(B_o - B_{oi})}{B_{oi}} + \frac{(c_w S_{wc} + c_f) \Delta p}{1 - S_{wc}} \right) \quad (3.15)$$

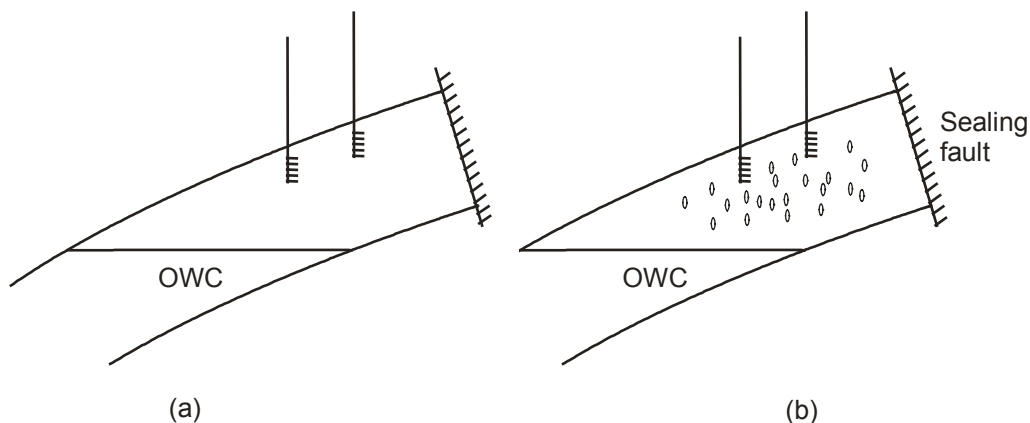


Fig. 3.2 Solution gas drive reservoir; (a) above the bubble point pressure; liquid oil, (b) below bubble point; oil plus liberated solution gas

The component describing the reduction in the hydrocarbon pore volume, due to the expansion of the connate water and reduction in pore volume, cannot be neglected for an undersaturated oil reservoir since the compressibilities c_w and c_f are generally of the same order of magnitude as the compressibility of the oil. The latter may be expressed as described in Chapter 2, sec. 6, as



$$c_o = \frac{(B_o - B_{oi})}{B_{oi} \Delta p}$$

and substituting this in equ. (3.15) gives

$$N_p B_o = NB_{oi} \left(c_o \frac{(c_w S_{wc} + c_f)}{1 - S_{wc}} \right) \Delta p \quad (3.16)$$

Since there are only two fluids in the reservoir, oil and connate water, then the sum of the fluid saturations must be 100% of the pore volume, or

$$S_o + S_{wc} = 1$$

and substituting the latter in equ. (3.16) gives the reduced form of material balance as

$$N_p B_o = NB_{oi} \left(\frac{c_o S_o + c_w S_{wc} + c_f}{1 - S_{wc}} \right) \Delta p \quad (3.17)$$

or

$$N_p B_o = NB_{oi} c_e \Delta p \quad (3.18)$$

in which

$$c_e = \frac{1}{1 - S_{wc}} (c_o S_o + c_w S_{wc} + c_f) \quad (3.19)$$

is the effective, saturation-weighted compressibility of the reservoir system. Since the saturations are conventionally expressed as fractions of the pore volume, dividing by $1 - S_{wc}$ expresses them as fractions of the hydrocarbon pore volume.

Thus the compressibility, as defined in equ. (3.19), must be used in conjunction with the hydrocarbon pore volume. Equation (3.18) illustrates how the material balance can be reduced to nothing more than the basic definition of compressibility, equ. (1.12), in which $N_p B_o = dV$, the reservoir production expressed as an underground withdrawal, and $NB_{oi} = V$ the initial hydrocarbon pore volume.

EXERCISE 3.1 SOLUTION GAS DRIVE; UNDERSATURATED OIL RESERVOIR

Determine the fractional oil recovery, during depletion down to bubble point pressure, for the reservoir whose PVT parameters are listed in table 2.4 and for which

$$\begin{aligned} c_w &= 3.0 \times 10^{-6} / \text{psi} & S_{wc} &= .20 \\ c_f &= 8.6 \times 10^{-6} / \text{psi} \end{aligned}$$

EXERCISE 3.1 SOLUTION

The data required from table 2.4 are

$$p_i = 4000 \text{ psi} \quad B_{oi} = 1.2417 \text{ rb/stb}$$



$$p_b = 3330 \text{ psi} \quad B_{ob} = 1.2511 \text{ rb/stb}$$

Therefore, the average compressibility of the undersaturated oil between initial and bubble point pressure is

$$c_o = \frac{B_{ob} - B_{oi}}{B_{oi} \Delta p} = \frac{1.2511 - 1.2417}{1.2417(4000 - 3330)} = 11.3 \times 10^{-6} / \text{psi}$$

The recovery at bubble point pressure can be calculated using equ. (3.18) as

$$\left. \frac{N_p}{N} \right|_{p_b} = \frac{B_{oi}}{B_{ob}} c_e \Delta p$$

where

$$\begin{aligned} c_e &= \frac{1}{.8} (11.3 \times 0.8 + 3 \times 0.2 + 8.6) \times 10^{-6} / \text{psi} \\ &= 22.8 \times 10^{-6} \end{aligned}$$

and therefore,

$$\text{Recovery} = \frac{1.2417}{1.2511} \times 22.8 \times 10^{-6} \times (4000 - 3330)$$

or 1.52% of the original oil in place. Considering that the 670 psi pressure drop represents about 17% of the initial, absolute pressure, the oil recovery is extremely low. This is because the effective compressibility is small providing the reservoir contains just liquid oil and water. The situation will, however, be quite different once the pressure has fallen below bubble point.

b) Below bubble point pressure (saturated oil)

Below the bubble point pressure gas will be liberated from the saturated oil and a free gas saturation will develop in the reservoir. To a first order of approximation the gas compressibility is $c_g \approx 1/p$, as described in Chapter 1, sec. 6. Therefore, using the data of exercise 3.1, the minimum value of the free gas phase compressibility will occur at the bubble point pressure and will be equal to $1/p_b = 1/3330 = 300 \times 10^{-6}/\text{psi}$. This is two orders of magnitude greater than the water compressibility and 35 times greater than the pore compressibility and, as a result, the latter two are usually neglected in the material balance equation. The manner in which the reservoir will now behave is illustrated by the following exercise.

EXERCISE 3.2 SOLUTION GAS DRIVE; BELOW BUBBLE POINT PRESSURE

The reservoir described in exercise 3.1 will be produced down to an abandonment pressure of 900 psia.

- 1) Determine an expression for the recovery at abandonment as a function of the cumulative gas oil ratio R_p .



What do you conclude from the nature of this relationship?

- 2) Derive an expression for the free gas saturation in the reservoir at abandonment pressure.

All PVT data may be taken from table 2.4.

EXERCISE 3.2 SOLUTION

- 1) For a solution gas drive reservoir, below the bubble point, the following are assumed
 - $m = 0$; no initial gascap
 - negligible water influx
 - the term $NB_{oi} \left(\frac{c_w S_{wc} + c_f}{1 - S_{wc}} \right) \Delta p$ is negligible once a significant free gas saturation develops in the reservoir.

Under these conditions the material balance equation can be simplified as

$$N_p (B_o + (R_p - R_s)B_g) = N ((B_o - B_{oi}) + (R_{si} - R_s)B_g) \quad (3.20)$$

underground = expansion of the oil plus
withdrawal originally dissolved gas

and the recovery factor at abandonment pressure of 900 psia is

$$(RF)_{900} = \frac{N_p}{N} \bigg|_{900 \text{ psi}} = \frac{(B_o - B_{oi}) + (R_{si} - R_s) B_g}{B_o + (R_p - R_s) B_g} \bigg|_{900 \text{ psi}}$$

in which all the PVT parameters B_o , R_s and B_g are evaluated at the abandonment pressure. Using the data in table 2.4, the recovery factor can be expressed as

$$\frac{N_p}{N} \bigg|_{900} = \frac{(1.0940 - 1.2417) + (510 - 122) \cdot 0.00339}{1.0940 + (R_p - 122) \cdot 0.00339}$$

which can further be reduced to

$$\frac{N_p}{N} \bigg|_{900} = \frac{344}{R_p + 201}$$

This clearly demonstrates that there is an inverse relationship between the oil recovery and the cumulative gas oil ratio R_p , as illustrated in fig. 3.3.

The conclusion to be drawn from the relationship is that, to obtain a high primary recovery, as much gas as possible should be kept in the reservoir, which requires that the cumulative gas oil ratio should be maintained as low as possible. By keeping the



gas in the reservoir the total reservoir system compressibility in the simple material balance

$$dV = cV \Delta p$$

will be greatly increased by the presence of the gas and the dV , which is the production, will be large for a given pressure drop.

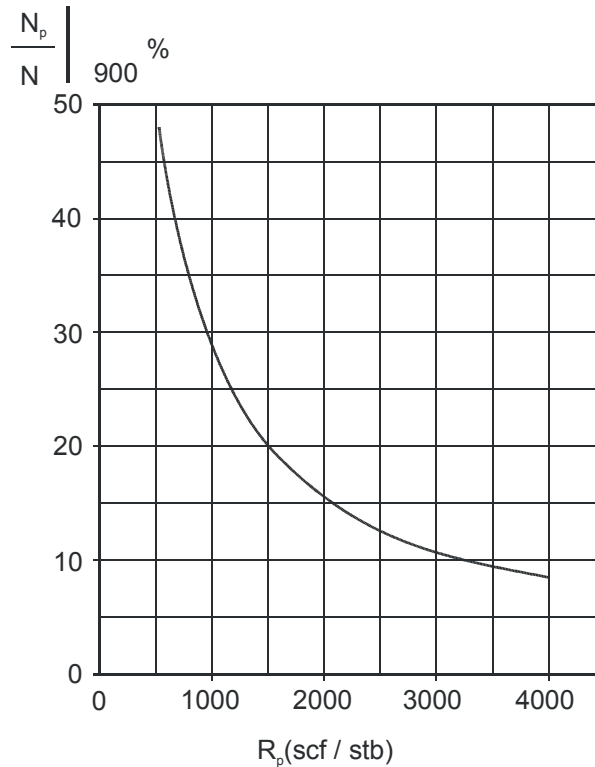


Fig. 3.3 Oil recovery, at 900 psia abandonment pressure (% STOILP), as a function of the cumulative GOR, R_p (Exercise 3.2)

- 2) The free gas saturation in the reservoir may be deduced in two ways, the most obvious being to consider the overall gas balance

$$\left[\begin{array}{c} \text{liberated} \\ \text{gas in the} \\ \text{reservoir} \end{array} \right] = \left[\begin{array}{c} \text{total} \\ \text{amount} \\ \text{of gas} \end{array} \right] - \left[\begin{array}{c} \text{gas} \\ \text{produced at} \\ \text{the surface} \end{array} \right] - \left[\begin{array}{c} \text{gas still} \\ \text{dissolved} \\ \text{in the oil} \end{array} \right]$$

which in terms of the basic PVT parameters can be evaluated at any reservoir pressure as

$$\text{liberated gas (rb)} = (NR_{si} - N_p R_p - (N - N_p) R_s) B_g$$

and expressing this as a saturation, which is conventionally required as a fraction of the pore volume, then

$$S_g = [N(R_{si} - R_s) - N_p(R_p - R_s)] B_g (1 - S_{wc}) / NB_{oi} \quad (3.21)$$

where $NB_{oi} / (1 - S_{wc}) = HCPV / (1 - S_{wc}) = \text{the pore volume.}$



A simpler and more direct method is to consider that

$$\left[\begin{array}{c} \text{liberated gas} \\ \text{in the} \\ \text{reservoir} \end{array} \right] = \left[\begin{array}{c} \text{initial total} \\ \text{volume of oil} \\ \text{in the reservoir} \end{array} \right] - \left[\begin{array}{c} \text{current oil} \\ \text{volume in} \\ \text{the reservoir} \end{array} \right]$$

$$\text{i.e. liberated gas} = NB_{oi} - (N - N_p) B_o \quad (\text{rb})$$

and therefore

$$S_g = (NB_{oi} - (N - N_p) B_o) (1 - S_{wc}) / NB_{oi}$$

or

$$S_g = 1 - \left(1 - \frac{N_p}{N} \right) \frac{B_o}{B_{oi}} (1 - S_{wc}) \quad (3.22)$$

which at abandonment pressure becomes

$$S_g = 1 - \left(1 - \frac{N_p}{N} \right) 0.88 \times 0.8$$

again showing that if the gas is kept in the reservoir so that S_g has a high value then N_p/N will be large, and vice versa.

Naturally equs. (3.21) and (3.22) are equatable through the material balance equ. (3.20).

Although the lesson of the last exercise is quite clear, the practical means of keeping the gas in the ground in a solution gas drive reservoir is not obvious. Once the free gas saturation in the reservoir exceeds the critical saturation for flow, then as noted already in Chapter 2, sec. 2, the gas will start to be produced in disproportionate quantities compared to the oil and, in the majority of cases, there is little that can be done to avert this situation during the primary production phase. Under very favourable conditions the oil and gas will separate with the latter moving structurally updip in the reservoir. This process of gravity segregation relies upon a high degree of structural relief and a favourable permeability to flow in the updip direction. Under more normal circumstances, the gas is prevented from moving towards the top of the structure by inhomogeneities in the reservoir and capillary trapping forces. Reducing a well's offtake rate or closing it in temporarily to allow gas-oil separation to occur may, under these circumstances, do little to reduce the producing gas oil ratio.

A typical producing history of a solution gas drive reservoir under primary producing conditions is shown in fig. 3.4.

As can be seen, the instantaneous or producing gas oil ratio R will greatly exceed R_{si} for pressures below bubble point and the same is true for the value of R_p . The pressure will initially decline rather sharply above bubble point because of the low compressibility of the reservoir system but this decline will be partially arrested once



free gas starts to accumulate. The primary recovery factor from such a reservoir is very low and will seldom exceed 30% of the oil in place.

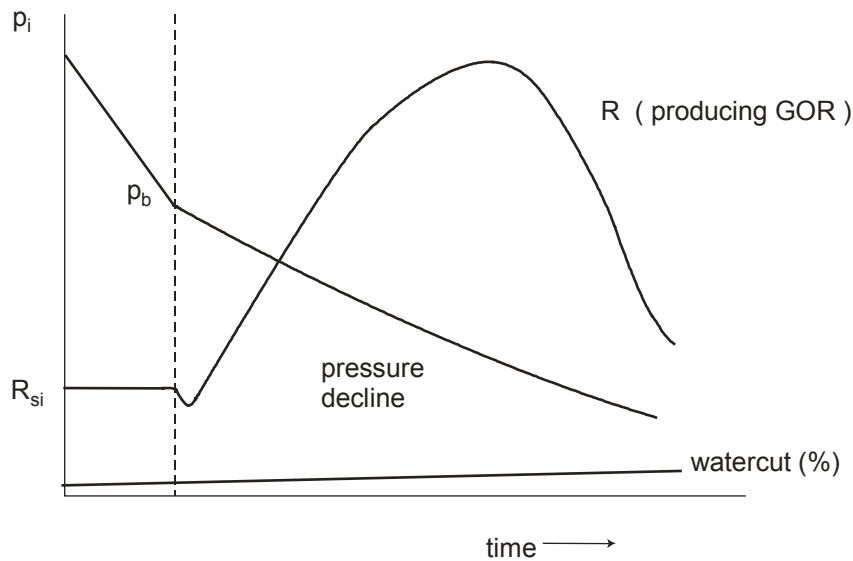


Fig. 3.4 Schematic of the production history of a solution gas drive reservoir

Two ways of enhancing the primary recovery are illustrated in fig. 3.5. The first of these methods, water injection, is usually aimed at maintaining the pressure above bubble point, or above the pressure at which the gas saturation exceeds the critical value at which the gas becomes mobile. The unfortunate consequences of starting to inject water below bubble point pressure are illustrated in exercise 3.3.

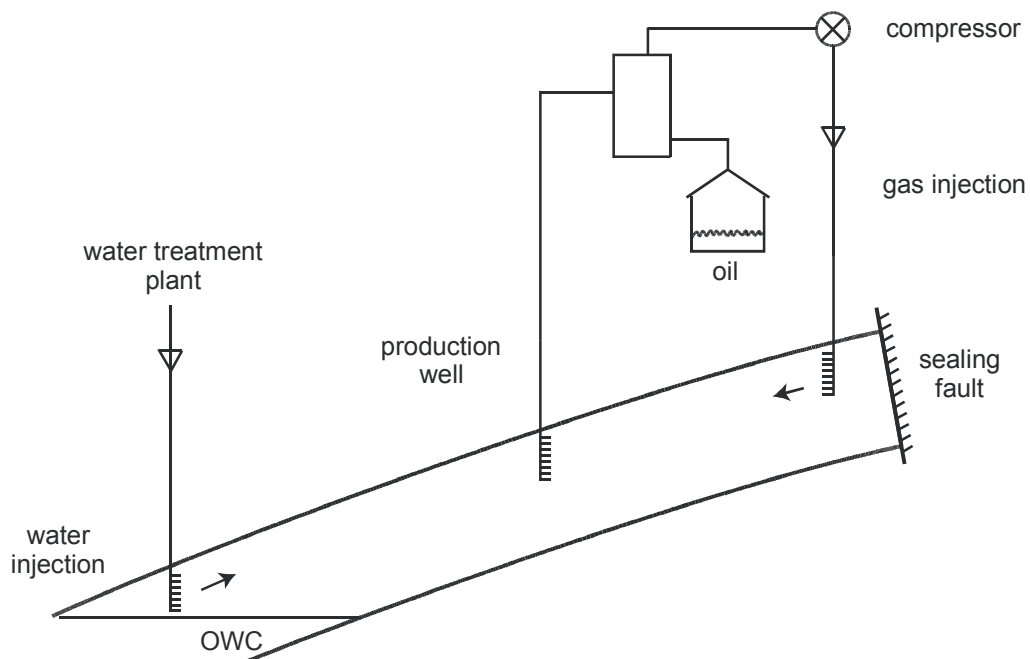


Fig. 3.5 Illustrating two ways in which the primary recovery can be enhanced; by downdip water injection and updip injection of the separated solution gas

EXERCISE 3.3 WATER INJECTION BELOW BUBBLE POINT PRESSURE

It is planned to initiate a water injection scheme in the reservoir whose PVT properties are defined in table 2.4. The intention is to maintain pressure at the level of 2700 psia ($p_b = 3330$ psia). If the current producing gas oil ratio of the field (R) is 3000 scf/stb, what will be the initial water injection rate required to produce 10,000 stb/d of oil.

EXERCISE 3.3 SOLUTION

To maintain pressure at 2700 psia the total underground withdrawal at the producing end of a reservoir block must equal the water injection rate at the injection end of the block. The total withdrawal associated with 1 stb of oil is

$$B_o + (R - R_s)B_g \quad \text{rb}$$

which, evaluating at 2700 psia, using the PVT data in table 2.4, is

$$\begin{aligned} &1.2022 + (3000 - 401) 0.00107 \\ &= 4.0 \text{ rb} \end{aligned}$$

Therefore, to produce 10,000 stb/d oil, an initial injection rate of 40,000 rb/d of water will be required, 70% of which will be needed to displace the liberated gas. If the injection had been started at, or above bubble point pressure, a maximum injection rate of only 12,500 b/d of water would have been required.

The mechanics of water injection are described in Chapter 10, including methods of calculating the recovery factor. One of the advantages in this secondary recovery process is that if the displacement is maintained at, or just below, bubble point pressure the producing gas oil ratio is constant and approximately equal to R_{si} .

If the gas quantities are sufficiently large it is easier, under these circumstances, to enter into a gas sales contract in which gas rates are usually specified by the customer at a plateau level. Conversely, there are obvious difficulties attached to entering such a contract with a gas oil ratio profile as shown in fig. 3.4. In such cases difficulties are frequently encountered in disposing of all the gas. Some portion of it may be sold under a fixed contract agreement but the remainder, which is frequently unpredictable in quantity, presents problems. In the "old days" (prior to the 1973–energy crisis) a lot of this excess gas, which could not conveniently be used as a local fuel supply, was flared. Even as late as the end of 1973 it was estimated that some 11% of the world's total daily gas production was flared. Today, regulations concerning gas disposal are more stringent and in many cases operators are obliged to re-inject excess gas back into the reservoir as shown in fig. 3.5. The gas is separated from the oil at high pressure and injected at a structurally high point thus forming a secondary gas cap. The oil production is taken from downdip in the reservoir thus allowing the high compressibility gas to expand and displace an equivalent amount of oil towards the



producing wells. This demonstrates a method of keeping as much gas in the reservoir as possible where it can serve its most useful purpose, as suggested in exercise 3.2.

The economic success of both water and solution gas injection depends upon the additional recovery obtained as a result of the projects. The present day value of the additional oil recovery must be greater than the cost of the injection wells, surface treatment facilities (mainly for water) and compressor costs (mainly for gas). In many cases, for small reservoirs, injection of water or gas is not economically viable and the solution gas drive process must be allowed to run its full course resulting in low oil recovery factors.

3.6 GASCAP DRIVE

A typical gascap drive reservoir is shown in fig. 3.6. Under initial conditions the oil at the gas oil contact must be at saturation or bubble point pressure. The oil further downdip becomes progressively less saturated at the higher pressure and temperature. Generally this effect is relatively small and reservoirs can be described using uniform PVT properties, as will be assumed in this text. There are exceptions, however, one of the most remarkable being the Brent field in the North Sea⁶ in which at the gas oil contact the oil has a saturation pressure of 5750 psi and a solution gas oil ratio of 2000 scf/stb, while at the oil water contact, some 500 feet deeper, the saturation pressure and solution gas oil ratio are 4000 psi and 1200 scf/stb, respectively. Such extremes are rarely encountered and in the case of the Brent field the anomaly is attributed to gravity segregation of the lighter hydrocarbon components.

For a reservoir in which gascap drive is the predominant mechanism it is still assumed that the natural water influx is negligible ($W_e = 0$) and, in the presence of so much high compressibility gas, that the effect of water and pore compressibilities is also negligible. Under these circumstances, the material balance equation, (3.7), can be written as

$$N_p (B_o + (R_p - R_s)B_g) = NB_{oi} \left[\frac{(B_o - B_{oi}) + (R_{si} - R_s)B_g}{B_{oi}} + m \left(\frac{B_g}{B_{gi}} - 1 \right) \right] \quad (3.23)$$

in which the right hand side contains the term describing the expansion of the oil plus originally dissolved gas, since the solution gas drive mechanism is still active in the oil column, together with the term for the expansion of the gascap gas. Equation (3.23) is rather cumbersome and does not provide any kind of clear picture of the principles involved in the gascap drive mechanism. A better understanding of the situation can be gained by using the technique of Havlena and Odeh, described in sec. 3.3, for which the material balance, equ. (3.12), can be reduced to the form

$$F = N (E_o + mE_g) \quad (3.24)$$



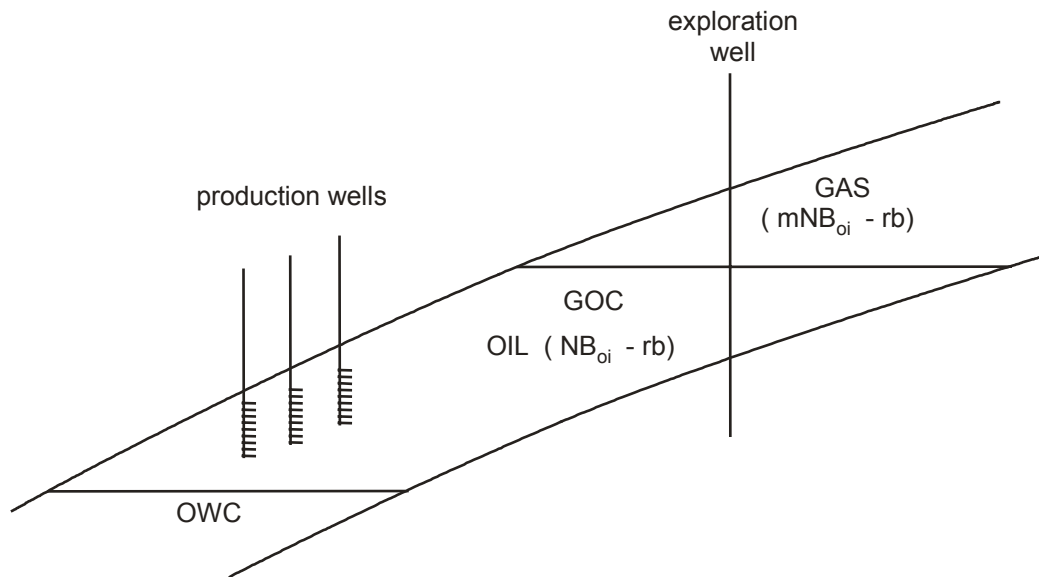


Fig. 3.6 Typical gas drive reservoir

The way in which this equation can be used depends on the unknown quantities. For a gascap reservoir the least certain parameter in equ. (3.24) is very often m , the ratio of the initial hydrocarbon pore volume of the gascap to that of the oil column. For instance, in the reservoir depicted in fig. 3.6, the exploration well penetrated the gascap establishing the level of the gas oil contact. Thereafter, no further wells penetrated the gascap since it is not the intention to produce this gas but rather to let it expand and displace oil towards the producing wells, which are spaced in rows further downdip. As a result there is uncertainty about the position of the sealing fault and hence in the value of m . The value of N , however, is fairly well defined from information obtained from the producing wells. Under these circumstances the best way to interpret equ. (3.24) is to plot F as a function of $(E_o + mE_g)$ for an assumed value of m . If the correct value has been chosen then the resulting plot should be a straight line passing through the origin with slope N , as shown in fig. 3.7. If the value of m selected is too small or too large, the plot will deviate above or below the line, respectively.

In making this plot F can readily be calculated, at various times, as a function of the production terms N_p and R_p , and the PVT parameters for the current pressure, the latter being also required to determine E_o and E_g . Alternatively, if N is unknown and m known with a greater degree of certainty, then N can be obtained as the slope of the straight line.

One advantage of this particular interpretation is that the straight line must pass through the origin which therefore acts as a control point.

EXERCISE 3.4 GASCAP DRIVE

The gascap reservoir shown in fig. 3.6 is estimated, from volumetric calculations, to have had an initial oil volume N of 115×10^6 stb. The cumulative oil production



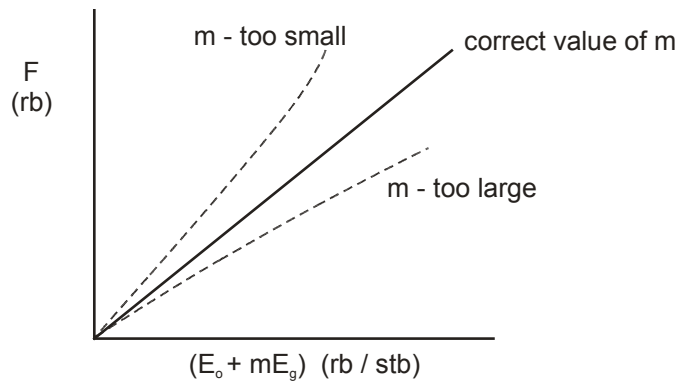


Fig. 3.7 (a) Graphical method of interpretation of the material balance equation to determine the size of the gascap (Havlena and Odeh)

N_p and cumulative gas oil ratio R_p are listed in table 3.1, as functions of the average reservoir pressure, over the first few years of production. (Also listed are the relevant PVT data, again taken from table 2.4, under the assumption that, for this particular application, $p_i = p_b = 3330$ psia).

Pressure psia	N_p MMstb	R_p scf/stb	B_o rb/stb	R_s scf/stb	B_g rb/scf
3330 ($p_i = p_b$)			1.2511	510	.00087
3150	3.295	1050	1.2353	477	.00092
3000	5.903	1060	1.2222	450	.00096
2850	8.852	1160	1.2122	425	.00101
2700	11.503	1235	1.2022	401	.00107
2550	14.513	1265	1.1922	375	.00113
2400	17.730	1300	1.1822	352	.00120

TABLE 3.1

The size of the gascap is uncertain with the best estimate, based on geological information, giving the value of $m = 0.4$. Is this figure confirmed by the production and pressure history? If not, what is the correct value of m ?

EXERCISE 3.4 SOLUTION

Using the technique of Havlena and Odeh the material balance for a gascap drive reservoir can be expressed as

$$F = N (E_o + mE_g) \quad (3.24)$$

where F , E_o and E_g are defined in equs. (3.8 – 10). The values of these parameters, based on the production, pressure and PVT data of table 3.1, are listed in table 3.2.



Pressure psia	F MM rb	E _o rb/stb	E _g rb/stb	m = .4	E _o + mE _g m = .5	m = .6
3330 (p _i)						
3150	5.807	.01456	.07190	.0433	.0505	.0577
3000	10.671	.02870	.12942	.0805	.0934	.1064
2850	17.302	.04695	.20133	.1275	.1476	.1677
2700	24.094	.06773	.28761	.1828	.2115	.2403
2550	31.898	.09365	.37389	.2432	.2806	.3180
2400	41.130	.12070	.47456	.3105	.3580	.4054

TABLE 3.2

The theoretical straight line for this problem can be drawn in advance as the line which, passes through the origin and has a slope of 115×10^6 stb, fig. 3.7 (b). When the plot is made of the data in table 3.2 for the value of $m = 0.4$, the points lie above the required line indicating that this value of m is too small. This procedure has been repeated for values of $m = 0.5$ and 0.6 and, as can be seen in fig. 3.7 (b), the plot for $m = 0.5$ coincides with the required straight line. Application of this technique relies critically upon the fact that N is known. Otherwise all three plots in fig. 3.7 (b), could be interpreted as straight lines, although the plots for $m = .4$ and $.6$ do have slight upward and downward curvature, respectively. Therefore, if there is uncertainty in the value of N , the three plots could be interpreted as

$$\begin{aligned} m &= 0.4 & N &= 132 \times 10^6 \text{ stb} \\ m &= 0.5 & N &= 114 \times 10^6 \text{ stb} \\ m &= 0.6 & N &= 101 \times 10^6 \text{ stb} \end{aligned}$$

If there is uncertainty in both the value of N and m then Havlena and Odeh suggest that equ. (3.24) should be re-expressed as

$$\frac{F}{E_o} = N + mN \frac{E_g}{E_o}$$

a plot of F/E_o versus E_g/E_o should then be linear with intercept N (when $E_g/E_o = 0$) and slope mN . Thus for the data given in tables 3.1 and 3.2

Pressure psia	F/E _o stb	E _g /E _o
3330 (p _i)		
3150	398.8×10^6	4.938
3000	371.8	4.509
2850	368.5	4.288
2700	355.7	4.246
2550	340.6	3.992
2400	340.8	3.932

TABLE 3.3



The plot of F/E_o versus E_g/E_o is shown in fig. 3.7 (c), drawn over a limited range of each variable. The least squares fit for the six data points is the solid line which can be expressed by the equation

$$\frac{F}{E_o} = (108.9 + 58.8 \frac{E_g}{E_o}) \times 10^6 \text{ stb}$$

and therefore according to this interpretation

$$N = 108.9 \times 10^6 \text{ stb, and } m = 0.54$$

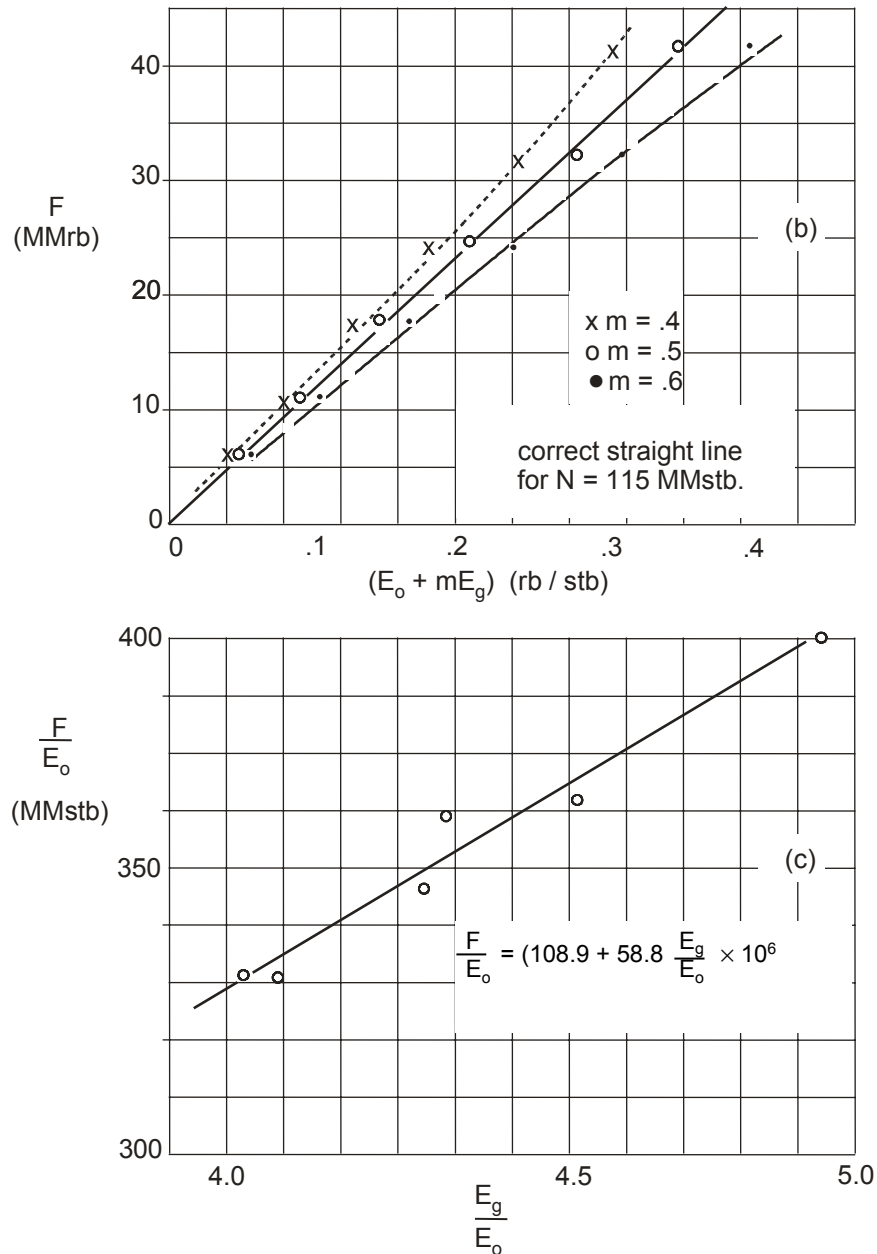


Fig. 3.7 (b) and (c); alternative graphical methods for determining m and N (according to the technique of Havlena and Odeh)



Both methods tend to confirm that the volumetric estimate of the oil in place is probably correct, within about 6%, and the gascap size is between $m = 0.5$ and 0.54 . With the slight scatter in the production data it is not meaningful to try and state these figures with any greater accuracy.

This estimate is made after the production of 17.7 million stb of oil or 15% recovery. As more production data become available the estimates of N and m can be revised.

The pressure and production history of a typical gascap drive reservoir, under primary recovery conditions, are shown in fig. 3.8.

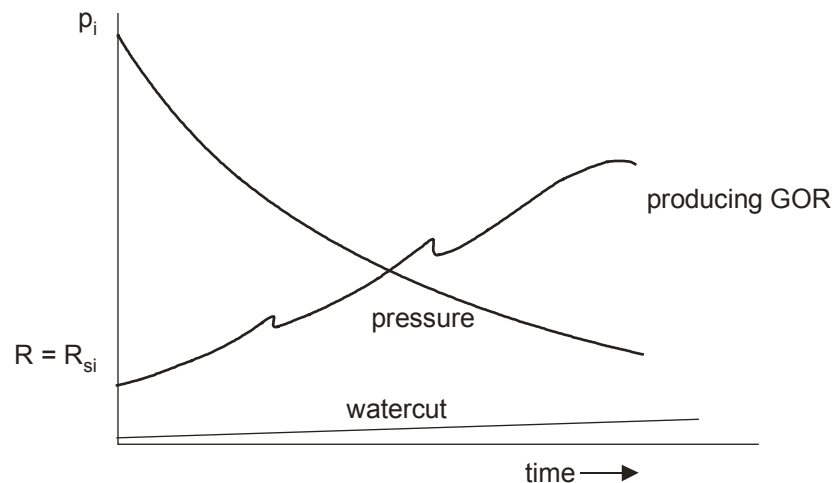


Fig. 3.8 Schematic of the production history of a typical gascap drive reservoir

Because of the gascap expansion, the pressure decline is less severe than for a solution gas drive reservoir and generally the oil recovery is greater, typically being in the range of 25–35 %, dependent on the size of the gascap. The peaks in the producing gas oil ratio curve are due to gas oil ratio (GOR) control being exercised. As the gascap expands the time will come when the updip wells start to produce gascap gas and the uppermost row of wells may have to be closed, both for the beneficial effect of keeping the gas in the reservoir and also to avoid gas disposal problems.

Just as described in sec. 3.5 for a solution gas drive reservoir, if the economics are favourable water and/or gas injection will enhance the ultimate recovery.

3.7 NATURAL WATER DRIVE

Natural water drive, as distinct from water injection, has already been qualitatively described, in Chapter 1, sec. 7, in connection with the gas material balance equation. The same principles apply when including the water influx in the general hydrocarbon reservoir material balance, equ. (3.7). A drop in the reservoir pressure, due to the production of fluids, causes the aquifer water to expand and flow into the reservoir.

Applying the compressibility definition to the aquifer, then

$$\text{Water Influx} = \text{Aquifer Compressibility} \times \text{Initial volume of water} \times \text{Pressure Drop}$$



or

$$W_e = (c_w + c_f) W_i \Delta p \quad (3.25)$$

in which the total aquifer compressibility is the direct sum of the water and pore compressibilities since the pore space is entirely saturated with water. The sum of c_w and c_f is usually very small, say $10^{-5}/\text{psi}$, therefore, unless the volume of water W_i is very large the influx into the reservoir will be relatively small and its influence as a drive mechanism will be negligible. If the aquifer is large, however, equ. (3.25) will be inadequate to describe the water influx. This is because the equation implies that the pressure drop Δp , which is in fact the pressure drop at the reservoir boundary, is instantaneously transmitted throughout the aquifer. This will be a reasonable assumption only if the dimensions of the aquifer are of the same order of magnitude as the reservoir itself. For a very large aquifer there will be a time lag between the pressure change in the reservoir and the full response of the aquifer. In this respect natural water drive is time dependent. If the reservoir fluids are produced too quickly, the aquifer will never have a chance to "catch up" and therefore the water influx, and hence the degree of pressure maintenance, will be smaller than if the reservoir were produced at a lower rate. To account for this time dependence in water influx calculations requires a knowledge of fluid flow equations and the subject will therefore be deferred until Chapter 9, in which a full description of the phenomenon is provided. For the moment, the simple equation (3.25) will be used to illustrate the influence of water influx in the material balance.

Using the technique of Havlena and Odeh (assuming that $B_w = 1$), the full material balance can be expressed as

$$F = N (E_o + mE_g + E_{f,w}) + W_e \quad (3.12)$$

in which the term $E_{f,w}$, equ. (3.11), can frequently be neglected when dealing with a water influx. This is not only for the usual reason that the water and pore compressibilities are small but also because a water influx helps to maintain the reservoir pressure and therefore, the Δp appearing in the $E_{f,w}$ term is reduced.

This is a point which should be checked at the start of any material balance calculation (refer exercise 9.2). If, in addition, the reservoir has no initial gascap then equ. (3.12) can be reduced to

$$F = NE_o + W_e \quad (3.26)$$

In attempting to use this equation to match the production and pressure history of a reservoir, the greatest uncertainty is always the determination of the water influx W_e . In fact, in order to calculate the influx the engineer is confronted with what is inherently the greatest uncertainty in the whole subject of reservoir engineering. The reason is that the calculation of W_e requires a mathematical model which itself relies on the knowledge of aquifer properties. These, however, are seldom measured since wells are not deliberately drilled into the aquifer to obtain such information. For instance, suppose the influx could be described using the simple model presented as equ. (3.25). Then, if the aquifer shape is radial, the water influx can be calculated as



$$W_e = (c_w + c_f) \pi (r_e^2 - r_o^2) f h \phi \Delta p \quad (3.27)$$

in which r_e and r_o are the radii of the aquifer and reservoir, respectively, and f is the fractional encroachment angle which is either $\Theta/2\pi$ or $\Theta/360^\circ$, depending on whether Θ is expressed in radians or degrees. It should be realised that the only term in equ. (3.27) which is known with any degree of certainty is π ! The remaining terms all carry a high degree of uncertainty. For instance, what is the correct value of r_e ? Is the aquifer continuous for 20 kilometers or is it truncated by faulting? What is the correct value of h , the average thickness of the aquifer or ϕ , the porosity? These can only be estimated, based on the values determined in the oil reservoir. For such reasons, building a correct aquifer model to match the production and pressure data of the reservoir is always done on a "try it and see" basis and even when a satisfactory model has been achieved it is seldom if ever, unique. Therefore, the most appropriate way of applying equ. (3.26) is by expressing it as

$$\frac{F}{E_o} = N + \frac{W_e}{E_o} \quad (3.28)$$

and plotting F/E_o , corresponding to the observed production, versus W_e/E_o , where W_e is calculated using an aquifer model such as equ. (3.27).

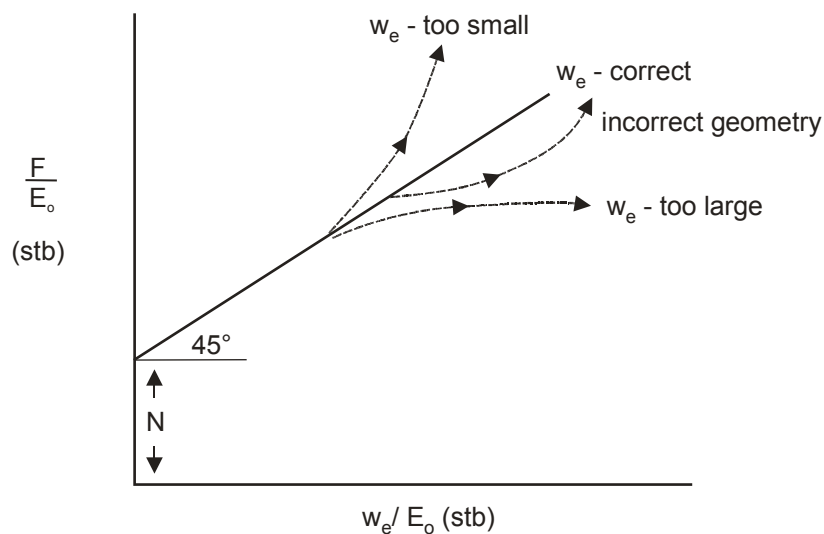


Fig. 3.9 Trial and error method of determining the correct aquifer model (Havlena and Odeh)

This model is linked to the reservoir by the pressure drop term Δp which is interpreted as the pressure drop at the original reservoir-aquifer boundary, and is normally assumed to be equal to the average pressure drop in the reservoir due to the production of fluids. If the aquifer model is incorrect, the plotted data points will deviate from the theoretical straight line which has a slope of 45° and intercept N , when $W_e/E_o = 0$, as shown in fig. 3.9.

The deviation labelled as being due to using the wrong geometry means that radial geometry has been assumed whereas linear geometry would probably be more appropriate. With radial geometry there is a larger body of water in close proximity to



the reservoir, for the same aquifer volume, than for a linear aquifer and, as a result, response of the radial aquifer is greater causing deviation below the theoretical straight line. Exercise 9.2 provides an example of this technique in which the aquifer model used for calculating W_e caters for time dependence.

Once a satisfactory aquifer model has been obtained by history matching, the same model can hopefully be used in predicting reservoir performance for any scheduled offtake policy. As already mentioned, however, there are so many uncertainties involved that the aquifer model is hardly ever unique and its validity should be continually checked as fresh production and pressure data become available.

If the reservoir has a gascap then equ. (3.12) has the form

$$F = N (E_o + mE_g) + W_e$$

which can alternatively be expressed as

$$\frac{F}{(E_o + mE_g)} = N + \frac{W_e}{(E_o + mE_g)} \quad (3.29)$$

in which it is assumed that both m and N are known.

By plotting $F/(E_o + mE_g)$ versus $W_e/(E_o + mE_g)$ the interpretation is similar to that shown in fig 3.9.

Equation (3.29) demonstrates how the technique of Havlena and Odeh can be applied to a combination drive reservoir in which there are three active mechanisms, solution gas drive, gascap drive and water drive.

The pressure and production history of an undersaturated reservoir under active water drive are shown in fig. 3.10. The pressure decline is relatively small due to the expansion of the aquifer water and from the producing gas oil ratio plot, it is evident that the pressure is being maintained above the saturation pressure. Recovery from water drive reservoirs can be very high, in excess of 50%, but just as in the case of the flooded out gas reservoir described in Chapter 1, sec. 7, residual oil will now be trapped behind the advancing water which can only be recovered by resorting to more advanced recovery methods, as described in Chapter 4, sec. 9.

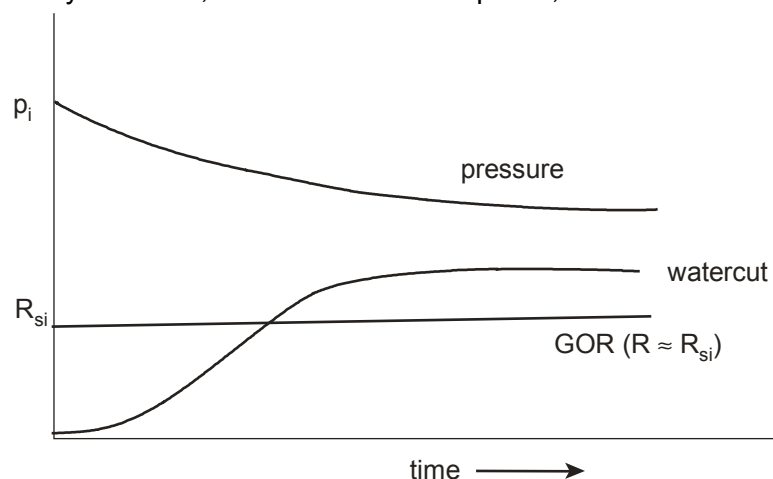


Fig. 3.10 Schematic of the production history of an undersaturated oil reservoir under strong natural water drive

3.8 COMPACTION DRIVE AND RELATED PORE COMPRESSIBILITY PHENOMENA

The withdrawal of liquid or gas from a reservoir results in a reduction in the fluid pressure and consequently an increase in the effective or grain pressure, the latter being defined in Chapter 1, sec. 3, as the difference between the overburden and fluid pressures. This increased pressure between the grains will cause the reservoir to compact and this in turn can lead to subsidence at the surface.

Various studies^{7,8,9,10} have shown that compaction depends only upon the difference between the vertically applied stress (overburden) and the internal stress (fluid pressure) and therefore, compaction can conveniently be measured in the laboratory by increasing the vertical stress on a rock sample while keeping the fluid pressure in the pores constant.

If V_b is the bulk volume of a rock sample of thickness h , then the uniaxial compaction

$$\Delta V_b/V_b = \Delta h/h$$

can best be determined in the laboratory using the triaxial compaction cell described by Teeuw¹¹, which is shown in fig. 3.11 (a).

The core sample, which is completely saturated with water, is contained in a cell which has permeable cap and base plates and a cylindrical, flexible sleeve surrounding it. Vertical stress is applied by means of a piston while the fluid pressure in the pores is maintained at one atmosphere. The pressure in the fluid surrounding the flexible sleeve can be increased independently so as to maintain the condition of

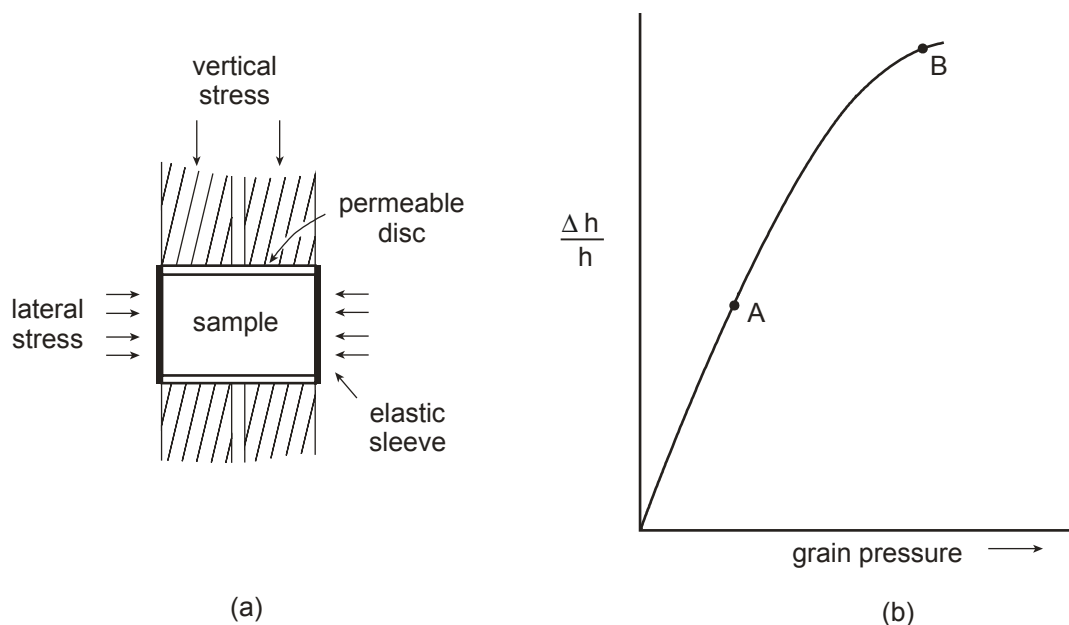


Fig. 3.11 (a) Triaxial compaction cell (Teeuw); (b) typical compaction curve



zero lateral strain on the sample. This pressure is continually adjusted so that any change in vertical thickness of the sample Δh is uniformly related to the measured water expelled from the porous rock.

If such an experiment were performed on an uncompacted sample of sand and the compaction $\Delta h/h$ plotted as a function of the applied vertical stress which, considering the fluid pressure is maintained at one atmosphere, is equivalent to the grain pressure, then the result would be as shown in fig. 3.11 (b). The slope of this curve, at any point, is

$$\frac{\Delta h}{h} / \Delta p = c_b \approx c_f \phi \quad \text{refer equ. (1.37)}$$

The characteristic shape of this compaction curve is intuitively what one would expect. At low grain pressures the compressibility of the uncompacted sample is very high since it is relatively easy to effect a closer packing of the grains at this stage. As the grain pressure increases, however, it becomes progressively more difficult to compact the sample further and the compressibility decreases. What is clear from such an experiment is that the bulk or pore compressibility of a reservoir is not constant but will continually change as fluids are withdrawn and the grain pressure increases.

Under normal hydrostatic conditions, since both the overburden and water pressures increase linearly with depth, then so too does the grain pressure which is the difference between the two. Thus a reservoir whose initial condition corresponds to point A will normally be buried at shallow depth, while a reservoir corresponding to point B will be buried deeper.

Compaction drive is the expulsion of reservoir fluids due to the dynamic reduction of the pore volume and will only be significant as a drive mechanism if the pore compressibility c_f is large. It therefore follows that such a drive mechanism will normally only provide a significant increase in the primary hydrocarbon recovery in shallow reservoirs. In parts of the Bachaquero field, Venezuela, as reported by Merle, et al¹², the compaction drive mechanism accounts for more than 50% of total oil recovery. This large reservoir dips between 1000–4000 ft. and has uniaxial compressibilities in excess of $100 \times 10^{-6}/\text{psi}$.

If the mechanics of reservoir compaction were as simple as described above, it would appear possible to derive a relationship between uniaxial compressibility and depth, for various types of typical reservoir rock, in an attempt to apply such a correlation universally. Unfortunately, the process of compaction is frequently irreversible which in turn implies that in-situ compressibility cannot be estimated in such a simple manner.

If the reservoir rock consists of well cemented grains in a rigid rock frame then the compaction, over a limited pressure range, will be approximately elastic and reversible. In loose unconsolidated sands, however, compaction is both inelastic and irreversible since upon each reloading cycle on such a sample, in a repeated loading experiment in a triaxial cell, it is possible for the individual grains to be packed in a different configuration than on the previous cycle and, in addition, some of the grains can suffer permanent mechanical deformation due to crushing. The effect of this inelastic



deformation in the reservoir is shown in fig. 3.12, which is taken from the paper of Merle et al.¹²

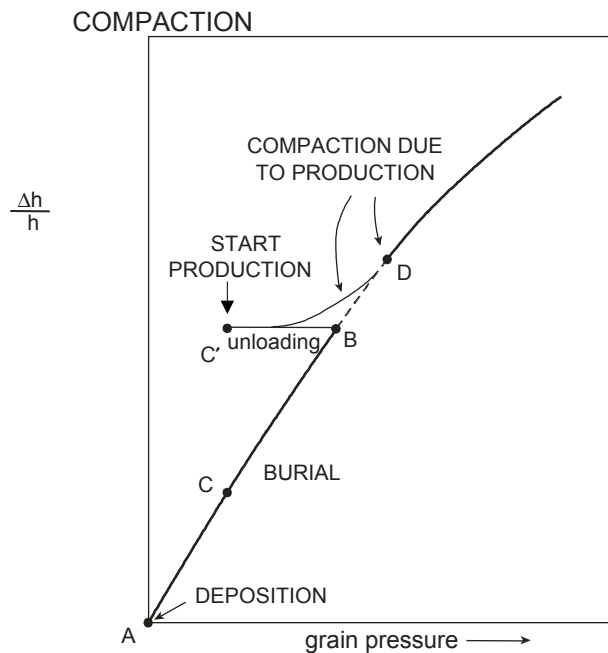


Fig. 3.12 Compaction curve illustrating the effect of the geological history of the reservoir on the value of the in-situ compressibility (after Merle)

When the reservoir sand is initially being deposited it is at point A on the compaction curve, fig. 3.12. Over geological times, as more and more material is deposited, the original sand becomes buried corresponding to point B, with grain pressure p_B . Following this normal deposition, events can occur which will reduce the grain pressure below p_B , such as:

- uplifting of the reservoir
- erosion of the surface layers above the reservoir
- overpressuring of the fluid in the reservoir.

As a result of one or more of these effects, in the extreme cases of either completely elastic or completely inelastic deformation of the rock during deposition, the reservoir in fig. 3.12 will be either at C or C', respectively, corresponding to the reduced grain pressure p_C . In the former case, for elastic deformation, if the reservoir is produced with an initial grain pressure p_C then the compaction will start immediately since the uniaxial compressibility at point C is finite. In the completely inelastic case, however, there will be a time lag between starting to produce the reservoir and the occurrence of any significant degree of compaction. This is because the uniaxial compressibility in this latter case is the tangent to the compaction curve at point C', which is extremely small. As shown in fig. 3.12, there will be very little compaction in the reservoir until sufficient fluids have been removed to increase the grain pressure to p_B which is the maximum grain pressure experienced by the reservoir in the past.



Compaction, and its associated effect of surface subsidence, will be much more pronounced for shallow, unconsolidated reservoirs than for the deeper, more competent sands. It is therefore necessary to experimentally determine the compressibility of shallow reservoir sands in order to estimate to what degree compaction will enhance the hydrocarbon recovery, and also, to enable the prediction of the resulting surface subsidence, which can cause serious problems if the surface location of the field is adjacent to the sea or a lake¹³.

Unfortunately, the deformation of unconsolidated sands is usually inelastic and this in turn leads to complications in relating laboratory measured compressibilities to the in-situ values in the reservoir. The nature of the problem can be appreciated by referring again to fig. 3.12. Suppose that both the grain and fluid pressures in a reservoir are normal so that under initial conditions the reservoir is at point B on the compaction curve. The process of cutting a core and raising it to the surface will cause unloading, which for a rock which deforms inelastically, will place the core at point C', which lies off the normal compaction curve. During the re-loading the horizontal path B-C' is not reversed, instead there is a mechanical hysteresis effect which means that the true compaction curve is not re-joined until point D, where $p_D > p_B$. As a result, the laboratory measured compressibility, determined as the slope of the line C'-D at pressure p_B , will be somewhat lower than the in-situ value, which is the slope of the normal curve at p_B . Thus, initial values of the in-situ compressibility are difficult to determine and usually require estimation by back extrapolation of laboratory values obtained for grain pressures in excess of p_D .

The above description of the various complications in estimating in-situ, uniaxial compressibility has been applied for the extreme case of a perfectly inelastic reservoir rock. Generally rock samples are neither perfectly elastic or inelastic but somewhere in between. Nevertheless, the same qualitative arguments apply and it is therefore not always meaningful to merely estimate in-situ compressibilities by reference to published charts for typical sandstones and limestones.

REFERENCES

- 1) Schilthuis, R.J., 1936. Active Oil and Reservoir Energy. Trans., AIME, 118: 33-52.
- 2) Amyx, J.W., Bass, D.M., and Whiting, R.L., 1960. Petroleum Reservoir Engineering - Physical Properties. McGraw-Hill: 448-472.
- 3) McCain, W.D., 1973. The Properties of Petroleum Fluids. Petroleum Publishing Company, Tulsa: 268-305.
- 4) Havlena, D. and Odeh, A.S., 1963. The Material Balance as an Equation of a Straight Line. J.Pet.Tech. August: 896-900. Trans., AIME, 228.
- 5) Havlena, D. and Odeh, A.S., 1964. The Material Balance as an Equation of a Straight Line. Part II - Field Cases. J.Pet.Tech. July: 815-822. Trans., AIME., 231.



- 6) Kingston, P.E. and Niko, H., 1975. Development Planning of the Brent Field. J.Pet. Tech. October: 1190-1198.
- 7) Geertsma, J., 1966. Problems of Rock Mechanics in Petroleum Production Engineering. Proc., 1st Cong. of the Intl. Soc. of Rock Mech., Lisbon. I, 585.
- 8) Geertsma, J., 1957. The Effect of Fluid Pressure Decline on Volumetric Changes of Porous Rocks. Trans., AIME, 210: 331-340.
- 9) van der Knaap, W., 1959. Non-linear Behaviour of Elastic Porous Media. Trans., AIME, 216: 179-187.
- 10) Biot, M.A., 1941. General Theory of Three Dimensional Consolidation. J.Appl. Phys., Vol.12: 155.
- 11) Teeuw, D., 1971. Prediction of Formation Compaction from Laboratory Compressibility Data. Soc. of Pet. Eng. J., September: 263-271.
- 12) Merle, H.A., Kentie, C.J.P., van Opstal, G.H.C. and Schneider, G.M.G., 1976. The Bachaquero Study - A Composite Analysis of the Behaviour of a Compaction Drive/ Solution Gas Drive Reservoir. J.Pet.Tech. September: 1107-1115.
- 13) Geertsma, J., 1973. Land Subsidence Above Compacting Reservoirs. J.Pet.Techn. June: 734-744.

